TRB Annual Meeting

Exploring Correlations between Vehicle Travels and Tropospheric Nitrogen Dioxide (NO2) Density among Florida Counties Impacted by COVID-19

--Manuscript Draft--

Full Title:	Exploring Correlations between Vehicle Travels and Tropospheric Nitrogen Dioxide (NO2) Density among Florida Counties Impacted by COVID-19
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Order of Authors:	Alican Karaer
	Nozhan Balafkan
	Michele Gazzea
	Reza Arghandeh
	Eren Erman Ozguven

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Alican Karaer
Graduate Student
Department of Civil & Environmental Engineering
FAMU & FSU College of Engineering
Tallahassee, FL 32310
Email: akaraer@fsu.edu

Nozhan Balafkan
External Consultant
Email: nozhan.balafkan@gmail.com

Michele Gazzea
Graduate Student
Department of Computer Science and Electrical Engineering
Western Norway University of Applied Sciences, Bergen, Norway
Email: michele.gazzea@hvl.no

Reza Arghandeh
Professor
Department of Computer Science and Electrical Engineering
Western Norway University of Applied Sciences Bergen, Norway
Email: arghandehr@gmail.com

Eren Erman Ozguven
Associate Professor
Department of Civil and Environmental Engineering
FAMU–FSU College of Engineering
2525 Pottsdamer Street, Tallahassee, FL 32310
Email: eozguven@fsu.edu

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Abstract

The COVID-19 outbreak and ensuing social distancing behaviors resulted in unprecedented changes in every aspect of daily life. The virus first hit the U.S. in March 2020, resulting in increased teleworking, closing university campuses and public entertainment places, and reducing many activities. This also led to a substantial decrease on every mode of transportation. As an important tropospheric trace gas and indicator of human activities, Nitrogen Dioxide (NO_2) density followed a similar pattern. This study investigates the relationship between these abrupt declines on traffic and NO_2 density in all the Florida Counties through March 2020. Daily tropospheric NO_2 concentrations were extracted from the Sentinel-5 Precursor satellite and Vehicle Mile Traveled (VMT) estimates were acquired from cell phone mobility records. Cross-correlation analysis was conducted with the Pearson's r statistic to analyze the similarities and associated time lag between 7-day moving averages of VMT and NO_2 density of 67 counties. Although such reduction is unprecedented for both data sets, results indicate a strong correlation and this correlation increases with the identification of a time lag between VMT and NO_2 density. Majority of the counties have no time lag between VMT and NO_2 density; however, a cluster of South Florida counties presents earlier decrease on NO_2 density compare to VMT, which can be attributed for a decrease in the industrial activities before the traffic reduction. Also, COVID-19 impacts on traffic and NO_2 density have started in the first half of the March 2020, earlier than the official stay-at-home orders.

Key words: *COVID-19, Nitrogen Dioxide (NO_2), Traffic, Vehicle Mile Traveled (VMT), Sentinel-5P, Remote Sensing*

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INTRODUCTION

The Coronavirus Disease 2019 (COVID-19) and ensuing social distancing behaviors have caused unprecedented changes in every aspect of modern life. Beyond the devastating impacts on human's health, daily life has changed tremendously to stop the spread of the novel Coronavirus (SARS-CoV-2). As of July 31st, more than 660 thousand people have died and more than 17 million people have exposed to this new virus around the world (*1*). Meanwhile, most global stock markets reported severe contractions in March 2020 (*2*), substantial increases were observed on unemployment, school campuses were closed, and a majority of remaining work activities has been performed remotely. To combat with this virus and analyze the impacts on various fields, the scientific community all over the world have come together in a truly unprecedented way.

In regard to the influence on the transportation and mobility, all the published literature states that the overall mobility has dropped significantly due to the spread of the virus (*3*, *4*). The success of a lockdown policy in China led most of the countries to apply similar countermeasures which mainly restricted mobility. An analysis carried out by INRIX (*5*) found a 60% reduction in commuting travels and 26% improvements on travel times in the City of Seattle using data from early March. In addition, commercial data providers such as Unacast Inc. *(9)* or StreetLight Data Inc *(10)* provided free access to their daily county level Vehicle Miles Traveled (VMT) data sets. The global mobility reports provided by Google (*6*) revealed a worldwide reduction in number of trips. As a result of these unprecedented changes on traffic demand, significant decreases were observed on traffic accidents, public transportation use and air pollution (*7*, *8*).

It has been known that air pollution is linked to different types of cardiovascular and respiratory diseases which leads to millions of premature deaths across the world every year. Indeed, a study estimated 360 premature death reduction in the U.S. due to the air quality improvements attributed from the decrease on traffic and electric use during the outbreak (*3*). However, long term estimations on the immediate impact of COVID-19 may not be an accurate assumption. Short-run impact on air quality can easily be outweighed in long-run effects of a decreasing clean energy innovation (*8*). Regardless, analyzing the relationship between traffic and air quality especially during these unprecedented times can help the cities with severe air pollution determine proper traffic restriction policies. For example, a study from India (*9*) suggested to continue with selected traffic restricted zones after observing a significant decrease on air pollution during the COVID-19 conditions.

Another study aspect of the immediate impact of COVID-19 on the air quality is the potential relationship on the spread of the virus. NASA has partnered with European Space Agency (ESA) to analyze how the shutdowns in response to the pandemic are changing the environment, especially the atmosphere, and determine what, if any, natural environmental phenomena might impact the spread of the pandemic (*10*). Preliminary results indicates that COVID-related impacts are harder to see on Sulfur Dioxide (SO_2) concentrations while significant decreases were observed on Nitrogen Dioxide (NO_2) density over China (*11*) and over the United States (*12*) using remote sensing data sources. Similarly, NOAA's Joint Polar Satellite System indicates an evident drop in NO_2 levels and particular matters (PMs) between March 2019 and 2020 (*13*). Further analyses are required to determine if the change is due to weather (i.e., wind) or a decrease in transportation.

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However, remotely sensed air pollution sources specifically point decreased NO_2 concentrations as an impact of decreased human activity due to COVID-19.

Nitrogen Dioxide (NO_2) takes a place among the highly reactive gases, known as Nitrogen Oxides (NO_x). Natural NO_2 emission sources such as biomass, nitrates of microbial reactions or atmospheric lightening process, have a very smaller contribution compare to the artificial NO_2 emission sources, which mainly include burning of fuel, industrial production and agricultural burning (*14*). NO_2 reacts with other chemicals in the air to form particulate matter and ozone, which are harmful to the respiratory system when inhaled. Additionally, NO_2 interacts with water, oxygen and other chemicals in the atmosphere to form acid rain, which harms sensitive ecosystems such as lakes and forests. Therefore, NO_2 is considered as one of the six criteria pollutants in National Ambient Air Quality Standards (NAAQS), which were established by U.S. Environmental Protection Agency (USEPA) through the Clean Air Act (*15*, *16*). NO_2 is continuously monitored across the U.S. so that those areas where the air quality does not meet the national standards can be identified and State Implementation Plans (SIP) can be developed to combat NO_2 pollution (*17*).

Given the preliminary results from space, transportation and environmental agencies, current literature **(TABLE 1)** also indicates a decrease on NO_2 levels due to COVID-19. Although other air pollutants vary based on the nature of implementation of the lockdown or additional pollution sources such as coal, a decrease in NO_2 has been seen in all studies. However, all these studies used surface-based sensors to acquire pollutant measurements. Although current regulatory compliance on air quality is performed mostly by surface sensors, the advancements in remote sensing methods and the availability of satellite data create new opportunities to use space level technologies to monitor every part of atmosphere rather than point observations on surface. The rapid development of the remote sensing monitoring methods of atmospheric satellite have gradually become critical technical means of global atmospheric monitoring (*14*). Thus, there is a need to evaluate the results with the satellite retrieved air pollution measurements.

As such, this study focuses on the impact of COVID-19 both on traffic and NO_2 densities investigating the relationship between the declines on traffic and NO_2 density in all the Florida Counties through March 2020. County level daily NO_2 levels are extracted from Sentinel-5P satellite of ESA, which expected to change the way the air quality is monitored today (*18*) and Vehicle Mile Traveled (VMT) estimates were acquired from cell phone mobility records. Population density, median income and the percentage of the households with no vehicle are also obtained for each county from the US Census's 2018 American Community Survey (*19*). Cross-correlation analysis was conducted with the Pearson's r statistic to analyze the similarities and associated time lag between 7-day moving averages of VMT and NO_2 density of 67 counties. Furthermore, the assessment of the relationship between the Vehicle Miles Travelled (VMT) and NO_2 can contribute to the research focusing on the causes of the observed NO_2 decrease all over the U.S. and the world.

The novelty of this study as follows:

- Evaluating the impact of COVID-19 and ensuing limited human activities on the daily NO_2 density levels across Florida with satellite technology.
- Providing an in-depth county level analysis on the potential correlation between the traffic and NO_2 concentration amid COVID-19.

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TABLE 1: Example studies on COVID-19 impact on air quality and their findings

Study	Location	Pollution Data	Traffic Data	Analysis Periods	Key Findings
Berman and Ebisu (*20*)	Continental US	Surface Sensors [OpenAQ (*21*)]	N/A	•Jan 8 – Mar 12 •Mar 13 – Apr 21 •2017 - 2019	•Average 25.5% NO_2 decrease •Rural area NO_2 decrease is smaller (16.5%) •Average 4.45% decrease on $PM_{2.5}$ •States with late or no stay-at-home order have increase (3.51%) on $PM_{2.5}$ but 11.26% decrease on early stay-at-home order states
Bekbulat et al. (*22*)	Continental US	Surface sensors (USEPA (*23*))	N/A	•After official stay-at-home order in each state •2010-2019	•Nationwide analysis for $PM_{2.5}$ and O_3 and 3 cities for NO_2 •~30% decrease on NO_2 but no significant decrease on $PM_{2.5}$ and O_3 •$PM_{2.5}$ and O_3 medians are decreasing 2 weeks before the stay-at-home orders but after it converges back with national averages. •For NO_2 in NYC, LA and Seattle decrease starts at 9th week before shutdown.
Jia (*24*)	Memphis, TN	Surface stations (USEPA (*23*))	StreetLight (*25*)	•Feb25-Mar24 •Mar25-May4 •2017-2019 April	•57% reduction on traffic •No significant reduction on $PM_{2.5,}$ O_3 during the lockdown compare to nor February neither same days last years •NO_2 measured from only 1 station and 12% increase observed on the mean compare to the previous years and no change compare to the pre-COVID •Estimation models indicates only wind speed is significant on estimating the NO_2 (Not lockdown, temperature, or precipitation)
Collivignarelli et al. (*26*)	Milan, Italy	Surface stations (Local Agency)	N/A	•Control [Feb7-20] •Partial lockdown [Mar9-22] •Total lockdown [Mar 23 – Apr 5]	•Total lockdown caused 61% and 74.5% reduction in city center on NO_2 and $NO_{x,}$ respectively. •47.4% and 48% decrease on PM_{10} and $PM_{2.5}$ •Substantial increase on O_3 due to minor NO concentration •Study periods are separated based on the temperature, humidity, wind, precipitation and solar irradiance.
Mahato et al. (*27*)	Delhi, India	Surface Stations (Local Agency)	N/A	• Pre Lockdown •After Lockdown	•+50% decrease on PM_{10} and $PM_{2.5}$ •52.68% and 30.35% decrease on NO_2 and CO respectively
Baldasano (*28*)	Barcelona and Madrid, Spain	Surface sensors (Local Agency)	Google Community Mobility Reports (*6*)	•March 2020 •March 2018/19	•62% and 50% NO_2 decrease in Madrid and Barcelona, respectively •75% reduction in traffic •NO_2 reduction in urban traffic area higher than urban and rural areas
Kerimray et al. (*29*)	Almaty, Kazakhstan	Surface stations (Local Agency)	N/A	•Mar 2-18 •Mar 19 –Apr 14	•35% NO_2 and 49% CO reduction due to the seasonal changes and traffic reduction. •Increase on SO_2 (7%) and O_3 (15%). •Seasonal reduction due to the reduce on coal use is more effective than traffic reduction. •21% reduction on $PM_{2.5}$ compare to last year with high variation.

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METHODOLOGY

Study Area

The state of Florida is selected as the test bed in this study. With a population of over 21 million, Florida is the third-most populous of the 50 states. Florida includes 67 counties where the populations of these counties range from 2 million populations to 8,000. **Figure 1** indicates population density of each county where those counties that host cities of Miami, Tampa, Orlando and Jacksonville appear to be the most populated ones. In addition, according to the Centers for Disease Control and Prevention (CDC), the older population (65+) and those with serious medical conditions such as lung disease, diabetes, liver disease, etc. are at a higher risk of COVID-19 infection (*30*). Hence, this becomes a serious issue in Florida since more than 20% of the total population in the state are 65 years and older (*19*). Therefore, the impact of COVID-19 on these populations becomes all the more confounding in light of their potential health and other safety concerns.

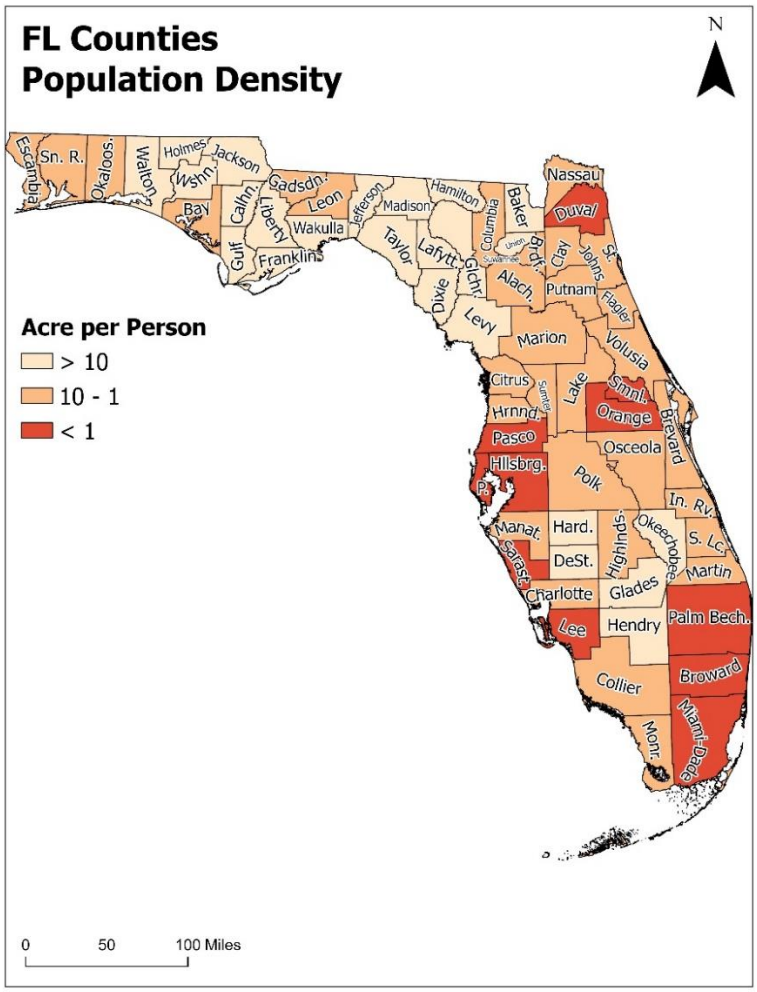

FIGURE 1: Study Area

Data Description

Generally, NO_2 pollutant measurement is performed in two different ways: ground-based and remote sensing-based methods. In ground-based methods, NO_2 is measured in air quality monitoring stations where NO_2 values are recorded several times a day using a set of sensors specifically designed to perform this task. While surface measurements serve as the tools for regulatory compliance, these measurements are limited to a few stations scattered over an area. In the remote sensing approach, on the other hand, NO_2 and other atmospheric trace gases are identified and measured by their signature spectral lines left in the reflected sun light on to the spectrometers. In the recent decades, advances in technology and reduced costs of building and launching satellites have made remote sensing a popular method for air quality monitoring; however, this method still lacks the required resolution to measure NO_2 and other atmospheric trace gases at the roadway level. A recent example of remotely monitoring NO_2 and other atmospheric trace gases is the Sentinel-5P, which has reached the city level resolution with 5×3.5 km (~3.1×2.2 mi.) pixel size (improved from 7×3.5 km (4.3×2.2 mi) in August 2019).

The Copernicus Sentinel-5 Precursor, known as Sentinel-5P, is a satellite in the series of Copernicus Sentinel satellites dedicated to atmospheric monitoring. This is a joint program between European Space Agency (ESA), the European Commission, the Netherlands Space Office, industry, data users, and scientists. Copernicus Sentinel-5P was successfully launched on October 13, 2017 with a planned mission of seven years. It aims to provide routine observations of atmospheric trace gases and aerosols. The TROPOspheric Monitoring Instrument (TROPOMI) is a spectral based device that uses the UV-Visible, NIR and SWIR bands to measure key atmospheric components such as Ozone (O_3), Sulphur Dioxide (SO_2), Carbon monoxide (CO), Methane (CH4), Formaldehyde (HCHO) additional to NO_2. Technology in TROPOMI has benefited from its predecessor instruments the Ozone Monitoring Instrument (OMI) and SCanning Imaging Absorption spectroMeter for Atmospheric CartograpHY (SCIAMACHY) (*31*).

For the purposes of this study, 34 files were downloaded from Sentinel-5P data hub (*32*). For each day, there were several files in data hub repository; however, an attempt was made to only download those files that had the total coverage over the State of Florida on a given date. There were 3 days (i.e., March 03, 08, and 19) on which there was no single file with the total coverage of Florida. Thus, three extra files were downloaded for these days. Next, data preparation was conducted with Python programming language. Careful attention was paid to ensure that the quality of data was not compromised. For this purpose, two different filters were applied to the input datasets for masking out faulty pixel values. The first filter masked out all pixels with NO_2 values below zero and the second filter masked out all pixels with quality-flag below 0.75, which is the recommended value in the user manual of the data set (*33*). The result was a set of pixels with NO_2 constriction values above zero and quality-flag value equal to or above 0.75. In the next step, using a county-level polygon geojson file (with a resolution of 20 m), each county was carefully extracted along with its associated NO_2 pixel values within the county. At this point, each county had a set of masked out and useful pixels. To estimate the average NO_2 over each county on a given day, we only used NO_2 values of the useful pixels. Also, we defined a parameter, called the coverage ratio, which indicates the fraction of useful pixels over a county to the total number of pixels in that county on a given day. The final product of this procedure was a list of counties with mean value of NO_2 and coverage ratio over that county for each day of the March 2020. This process is summarized in **Figure 2** with a flowchart. Also, NO_2 density measurements of 2

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example dates from March are illustrated in **Figure 3**. Note that the immediate impact of COVID is visible especially on the east side where Hillsborough and Pinellas counties are located. Since NO_2 density fluctuates between weekdays and weekends (*18*), 7-day moving averages are calculated for each county. This also helped interpolating the missing data due to the quality masking.

The daily Vehicle Miles Traveled (VMT) estimates for each county were extracted from StreetLight Data Inc. This commercial mobility data provider created a pro bono COVID-19 data set to help researchers and raise public awareness on social distancing. The data set includes daily VMTs of the U.S. counties beginning from March 1st. As a baseline, the average January 2020 VMTs are also provided for each county. StreetLight's VMT estimates are based on the data collected from the smart phone apps providing location-based services and GPS trackers on commercial fleets (*34*). In general, the daily VMT estimates were calculated by using the VMT of previous months as the baseline and multiplying the calculated mean trip length with the total number of trips. More details on their methodological and validation approach can be found in (*35*). To avoid the fluctuation between weekdays and weekend VMTs, 7-day moving averages were calculated for the month of March 2020.

In addition, population density, median income and the percentage of the households with no vehicle data were also obtained from the US Census's 2018 American Community Survey (*19*).

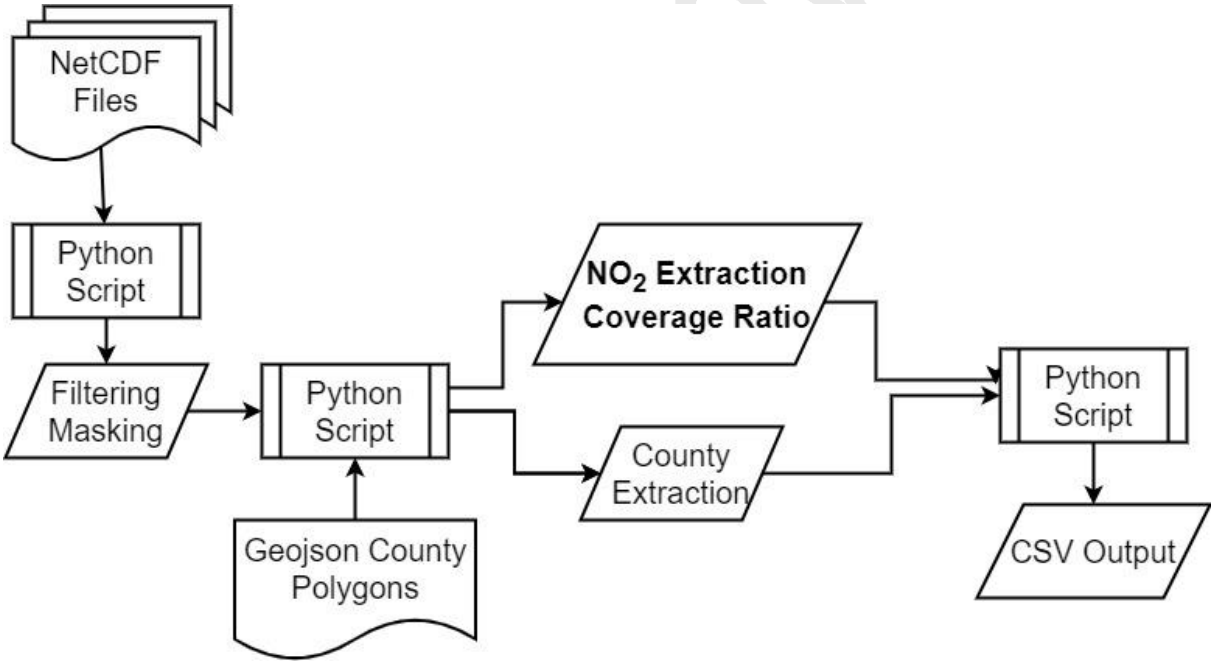

FIGURE 2: Flowchart for extracting daily NO_2 density for each county in Florida

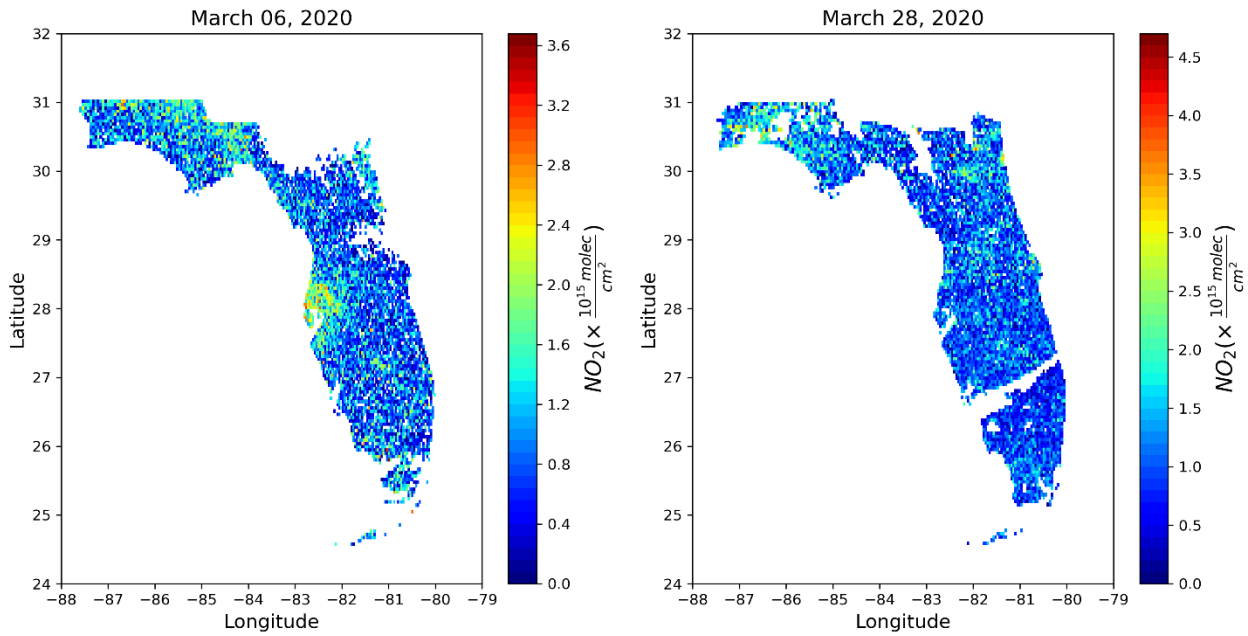

FIGURE 3: Examples of Sentinel-5P daily measurement for NO_2 densities over Florida

Data Analysis Approach

Cross-correlation is a well-known measure of similarity of two series as a function of the displacement (*36*). It measures the sliding dot product of two signals that turns a larger value when both signals are following a similar pattern and vice versa. Normalization of cross-correlation is a common approach to test the similarities of signals with different units (i.e., VMT and NO_2 densities), which can be interpreted as time dependent Pearson Correlation Coefficient (*37*). Mathematically, normalized cross correlation can be defined as:

$$\rho(\tau) = \frac{E[(X_t - \mu_X)(Y_{t+\tau} - \mu_Y)]}{\sigma_X \sigma_Y} \quad (1)$$

where the correlation between X_t and Y_t is calculated by sliding the Y_t signal with τ time step- or so-called lag. $\rho(\tau)$ represent the correlation coefficient on the lag τ and takes a place in [-1,1]. -1 indicates perfect anti correlation, and 1 indicating perfect correlation between the tested variables.

In this study, maximum cross correlation, $\rho(\tau^*)$ was calculated for 7-day moving averages of VMT and NO_2 density of each Florida county. τ^* can be interpreted as the time lag between VMT and NO_2 density through March 2020. If this value is negative, it indicates that the NO_2 decrease started after the VMT decrease, which shifts NO_2 to the left side on time providing a higher correlation. Also, $\rho(0)$ reveals the natural correlation coefficient between VMT and NO_2 density without any time sliding. Finally, a county with a positive τ^* value has observed the NO_2 decline earlier than the VMT decline and thus sliding its NO_2 density to the right side resulting in a higher correlation with VMT.

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To understand what effects the time lag between VMT and NO_2, other data such as population density, median income and the percentage of the households with no vehicle were also obtained for each county from the US Census's 2018 American Community Survey (*19*).

RESULTS

Short-Term Effects of COVID-19 in Daily Travel and NO_2 Emissions

After reporting a pneumonia cluster of unknown cause in Wuhan, China on January 5th, the novel Coronavirus (SARS-CoV-2) was first named on January 9th by the World Health Organization (WHO) (*38*). By the end of January 2020, a total of 9,976 cases had been reported in at least 21 countries, including the first confirmed case of Coronavirus Disease (COVID-19) in the U.S., confirmed on January 20th (*39*). In February, new cases with human-to-human interactions were being reported throughout the country almost every day (*40*). On February 29th, the first COVID-caused death in the U.S. was documented in a nursing home of Washington State (*40*). Eventually, Florida became the tenth state in the U.S. on March 1st to confirm the first COVID-19 cases in Manatee and Hillsborough Counties. Also, the first death in the East Coast of U.S. was reported on March 6th in Florida (*41*). However, there are strong evidences that the virus was causing unreported deaths since early February in all over the country (*41–44*).

Although a nation-wide lockdown has never been initiated in the U.S., social distancing behaviors such as teleworking and shifting to online teaching as well as the social distancing behaviors have led to a drastic decrease on overall mobility of the nation. StreetLight Data's pro bono Vehicle Mile Traveled (VMT) data for U.S. counties highlight this impact. **Figure 4(a)** indicates the 7-day moving average of VMT for all U.S. Counties included in the dataset whereas **Figure 4(b)** depicts the 67 Florida Counties along with 7-day moving averages of tropospheric NO_2 concentrations during March 2020. Also, January 2020 VMT averages and a generalized timeline of COVID-19 impact/respond is highlighted in **Figure 4**. It can be observed that, in the second half of the analyzing period, traffic demand decreased approximately 70% in both FL and the entire U.S. Although it varies based on the local government responses, the abrupt decline starts on March 13th, Friday with the national emergency and global outbreak announcements. Also, most of the universities started their spring break on this date and shifted to online delivery until the end of the semester. In Florida, stay-at-home order was first mandated 2 weeks later, on March 30 for only 3 counties: Miami-Dade, Broward, and Palm Beach, then on April 1st for the entire state. A study (*3*) analyzed the mobility differences between the early-policy states which put a stay-at-home policy in place by March 28, and the states with no policy by the same date. Both Florida and U.S. VMTs decreased compared to the January average on 17th which is parallel with the late or no policy states found in the study.

As aforementioned, NO_2 density over an area is strongly related with human activities. Therefore, it is not surprising that a parallel pattern was observed between 7-day moving averages of VMT and NO_2 density. To indicate the short-term effect of COVID-19, percentage decreases of VMT and NO_2 concentration were calculated based on their maximum and minimum values observed in March for each county:

$$[(max - \min)/max] * 100 \quad (2)$$

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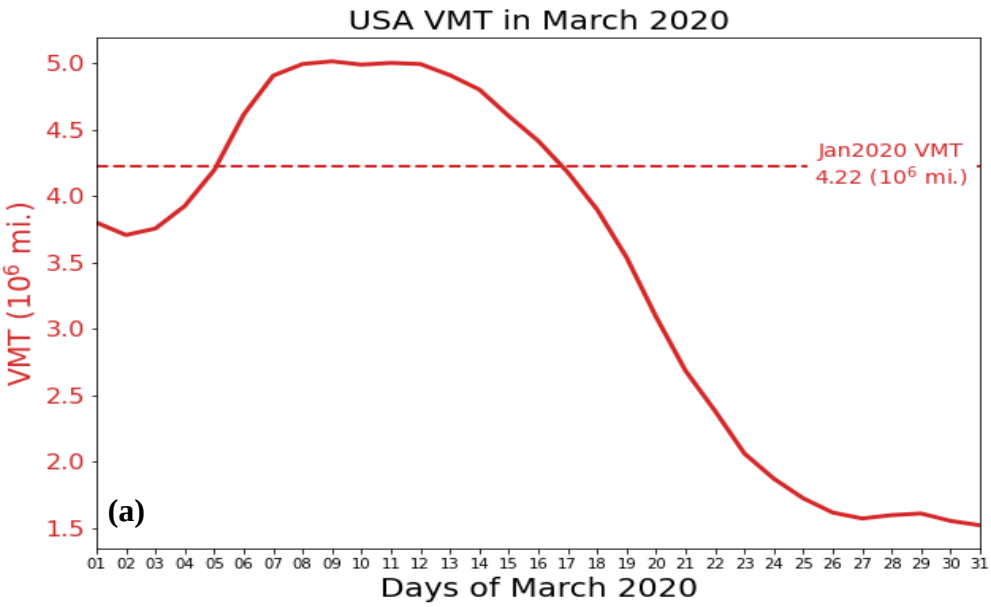
Among 67 Florida Counties, an average of 72.93% and 69.30% decrease were observed for NO_2 and VMT, respectively. The distributions of these percentage decreases are presented in **Figure 5**. Furthermore, the distributions for the dates in March when the maximum and minimum values were observed for VMT and NO_2 density of each county are presented in **Figure 6**. Results indicate that vehicle use in most of the counties started to decrease after the Friday, March 13th until the end of the month. This is more than 2 weeks before the official stay-at-home order was initiated in Florida on April 1st. This clearly indicates that social distancing behaviors have affected the traffic and air quality before the official policy enacted.

Despite the persistence on VMT dates, NO_2 density observations depict more variation on the dates when maximum and minimum values were observed. This can be attributed to different reasons including the characteristics of the county such as the level of urbanization (i.e., urban or rural), main industry, vehicle distribution, and favorable weather, which may prevent the NO_2 particles to get concentrated or change the impact of the COVID at a particular county. The relationship between VMT and NO_2 density is visible on the 7-day average diagrams; however, a cross-correlation-based dependency analysis can investigate the time lag, if there is any.



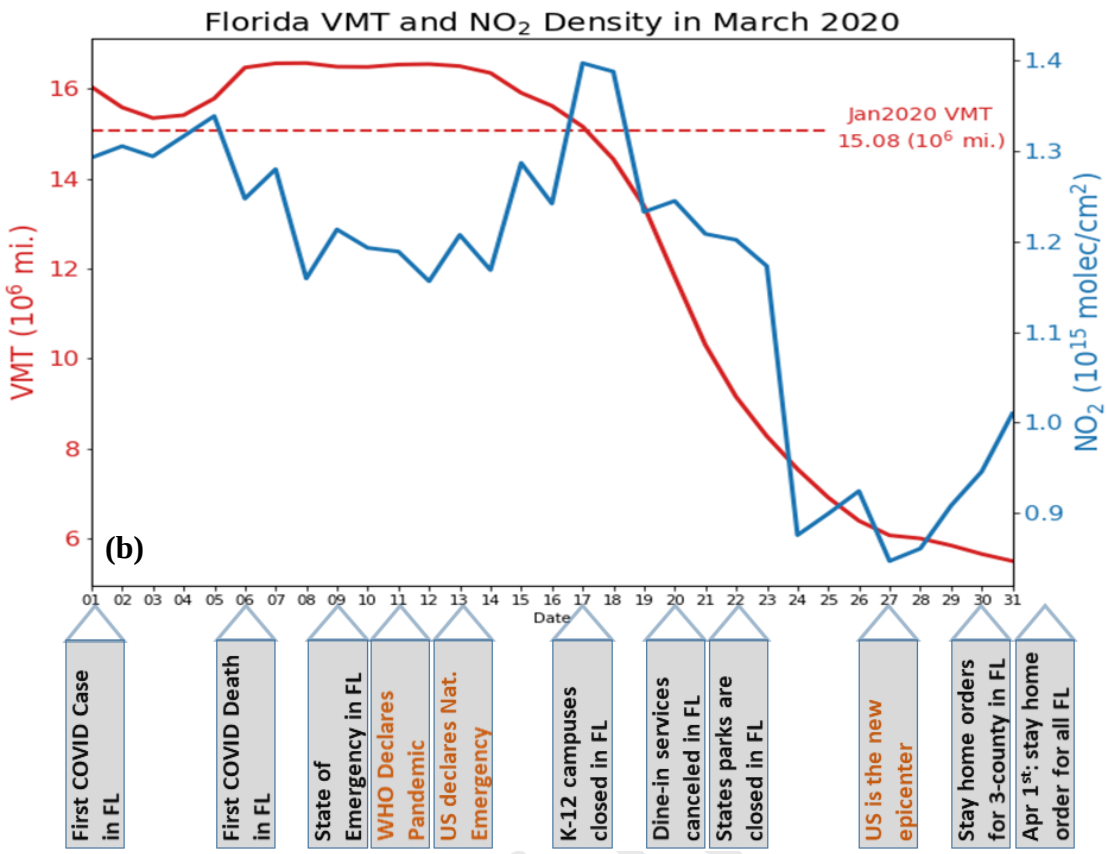


FIGURE 4: 7-Day moving averages of VMT in the U.S. (a) 7-day moving averages of VMT and NO2 density in Florida (b)

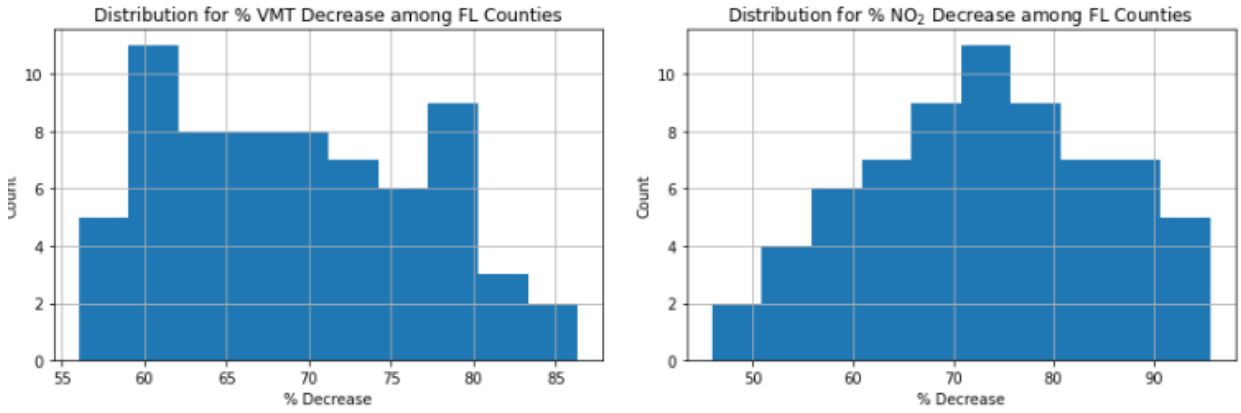


FIGURE 5: Distribution of % Decrease of VMT and NO_2 density among Florida counties

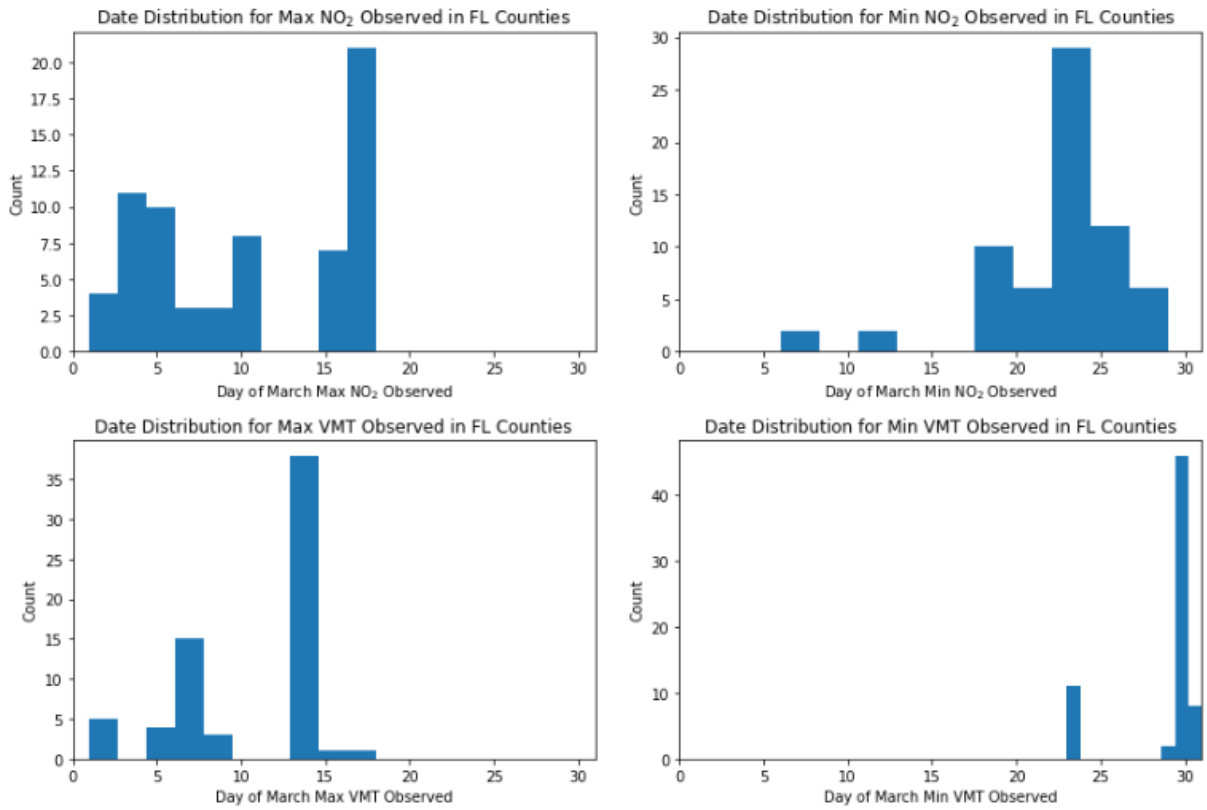


FIGURE 6: Date on which maximum and minimum values were observed for VMT and NO_2 density among Florida counties

Dependency Analysis on VMT and NO_2 Trends

Even with the abrupt decline, the immediate impact of COVID-19 in March 2020 indicates a similar behavior for both Vehicle Mile Traveled (VMT) and Nitrogen Dioxide (NO_2) density in Florida. To investigate this similarity and influencing factors, a cross-correlation analysis was conducted by calculating the correlation coefficients with 1-day displacement between 7-day moving averages of VMT and NO_2 density for each county. As mentioned in methodology section, the time lag is considered where the maximum correlation is calculated. A particular interest is to find the correlation and time lag between VMT and NO_2 with a focus on the socio-economic characteristics of each county.

Figure 7 depicts the positive, zero and negative time lags (τ^*) in each row for the counties with a high population density on the left column and a low population density on the right column. For Miami-Dade County, the decrease of NO_2 density started 5 days earlier than the decrease on VMT since the correlation increased from 0.65 to 0.80 by shifting the NO_2 density 5 days later. Miami-Dade County accommodates the highest number of population compared to other counties of Florida (*19*) and it has been hit the hardest by COVID-19. The county currently has approximately the 25% of all confirmed cases in the entire state. Although the first case was confirmed in

Hillsborough County, the VMT and NO_2 density started to decrease later than Miami-Dade County. Note that the lowest correlation was observed for the counties where NO_2 density started to decrease after VMT. This can be attributed to the non-traffic related factors such as weather, and commercial or industrial manufacturing.

To investigate the impact of economic status of the counties on the correlation between VMT and NO_2 density, counties with higher and lower median household income are compared in the **Figure 8.** Although the economic status of the area has a significant impact on the overall NO_2 density, no significant impact was observed on the correlation between VMT and NO_2 from the median income similar to the one with the population density.

The map for time lag is presented in **Figure 9** to illustrate the overall cross-correlation analysis results across Florida. In general, the counties in which the NO_2 decline was observed before the VMT decrease are clustered in the southern sections of the state whereas the counties where NO_2 decrease was observed after VMT are located in the center and northern parts of the state. Also, the distribution for the time lag indicates that most of the counties observed the VMT and NO_2 density decrease on the same days. Furthermore, the scatter graph for time lag (τ^*) and maximum correlation $\rho(\tau^*)$ indicates that the counties in which the NO_2 density decreased after the VMT (i.e., the counties with negative (τ^*) value) present a slightly less correlation compared to the counties with a positive (τ^*) value. Counties without a time lag ($\tau^* = 0$) indicates a higher variation on the correlation coefficient. This reveals that the effect of traffic on NO_2 density varies regardless of the time lag.

Additional to the population density and median income, registered heavy vehicle percentage and the percentage of the households with no vehicle for each county are also evaluated against the maximum correlation coefficient with different time lags. Surprisingly, a higher correlation was observed in the counties with higher percentage of households without a vehicle. This indicates that the increase on the households with no vehicle results in a higher correlation between VMT and NO_2. However, this can be negligible considering very low percentages on the zero-vehicle ownership.

To sum up, no significant impacts from the population, income and vehicle distribution of county characteristics were observed on the correlation between VMT and NO_2. Future studies shall include the weather and commercial industry information from each county so that additional impacts on the relationship between VMT and NO_2 density can be tested. Especially since the NO_2 production can also occur due to commercial manufacturing and food manufacturing, such information can help assess the time lags that occur in different counties.

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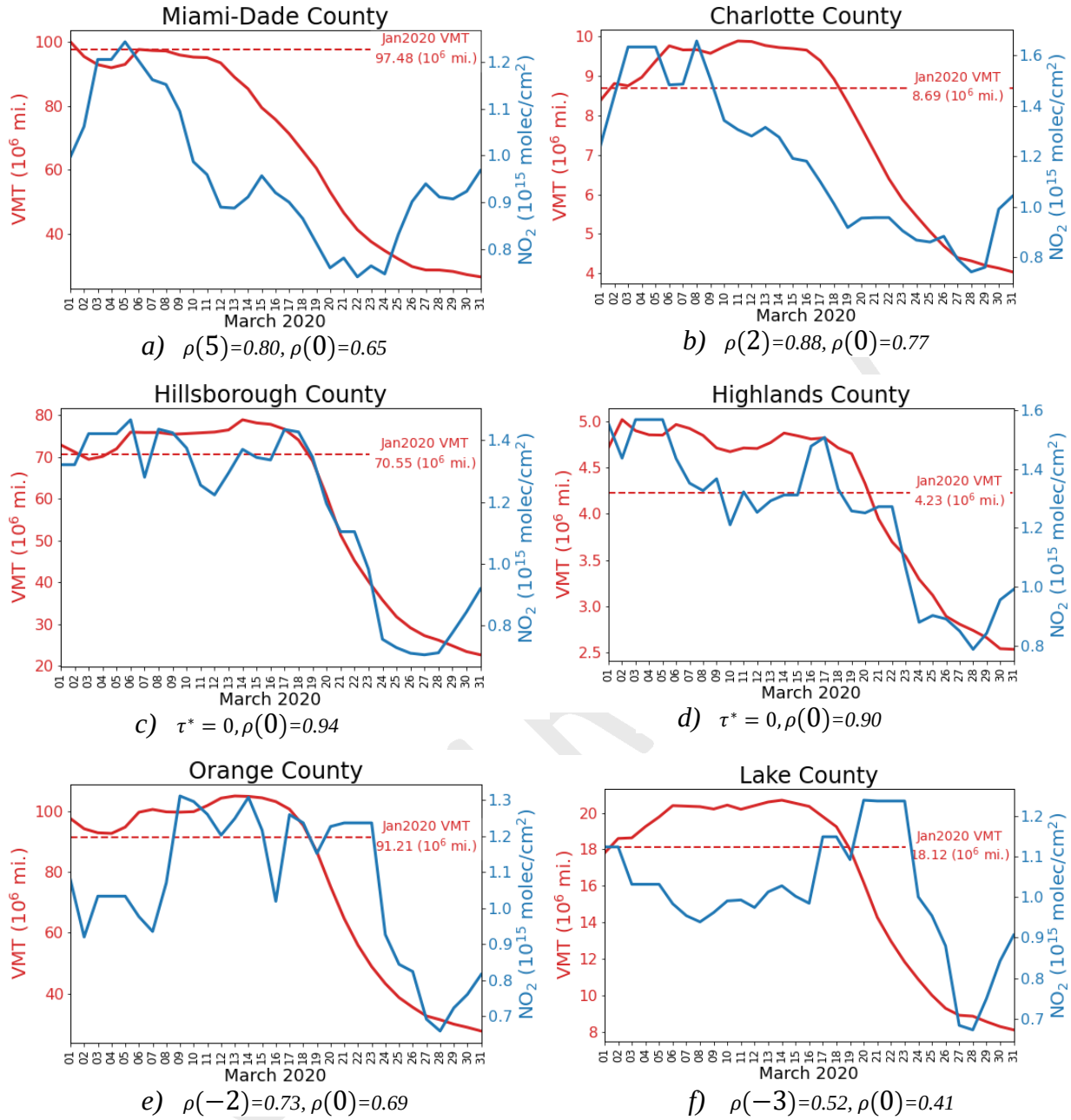
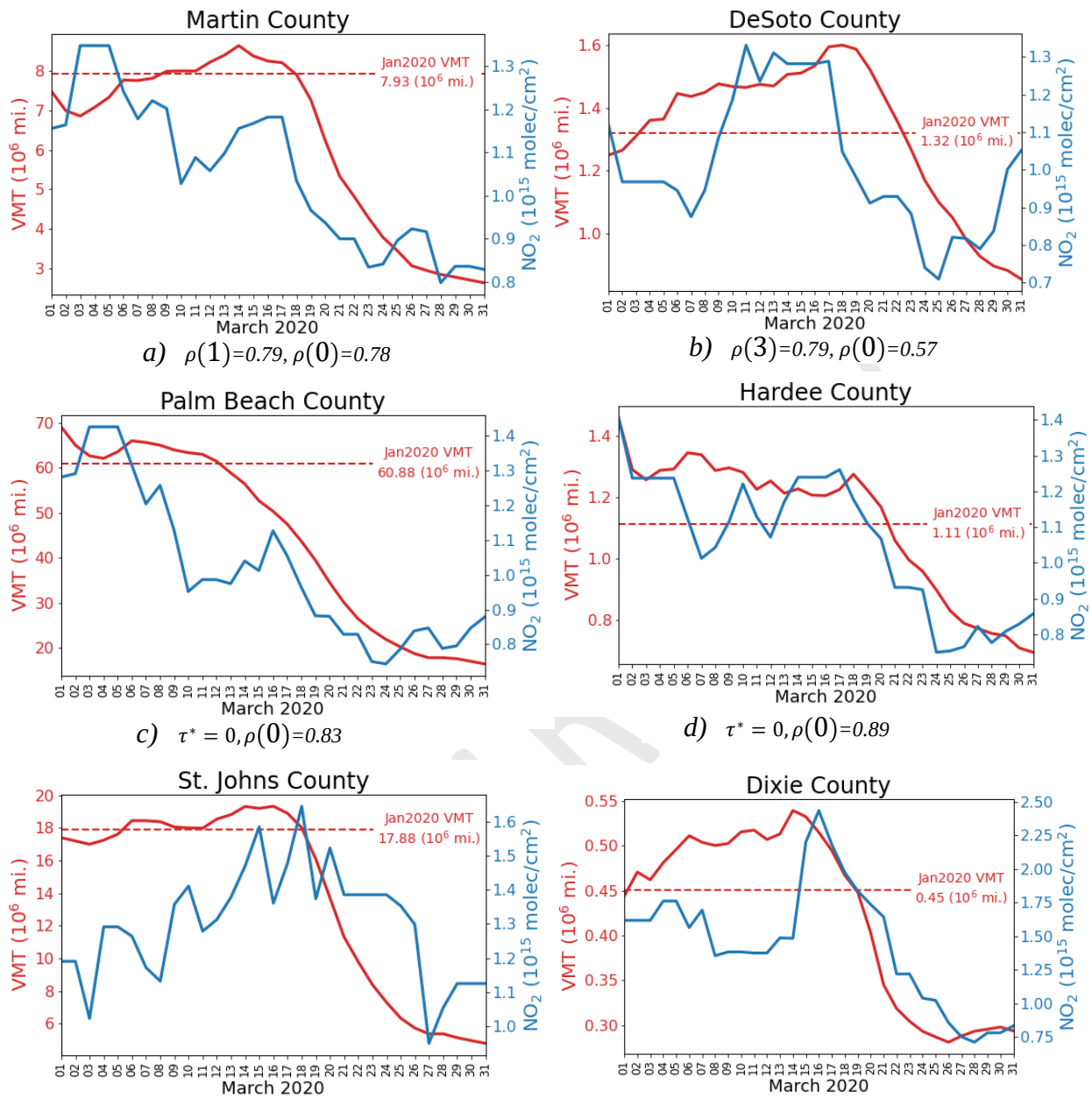


FIGURE 7: Different time lags (τ^*) and correlation $\rho(\tau^*)$ between 7-day moving averages of VMT and NO_2 in counties with high population density (left column <0.5 acre per person) and low population density (right column > 2 acre per person)



a) $\rho(1)$=*0.79*, $\rho(0)$=*0.78* *b)* $\rho(3)$=*0.79*, $\rho(0)$=*0.57*

c) $\tau^* = 0, \rho(0)$=*0.83* *d)* $\tau^* = 0, \rho(0)$=*0.89*

e) $\rho(-3)$=*0.39*, $\rho(0)$=*0.55* *f)* $\rho(-1)$=*0.73*, $\rho(0)$=*0.71*

FIGURE 8: Different time lags (τ^*) and correlation $\rho(\tau^*)$ between 7-day moving averages of VMT and NO_2 in counties with higher median income (left column > \$55,000) and lower median income (right column <\$40,000)

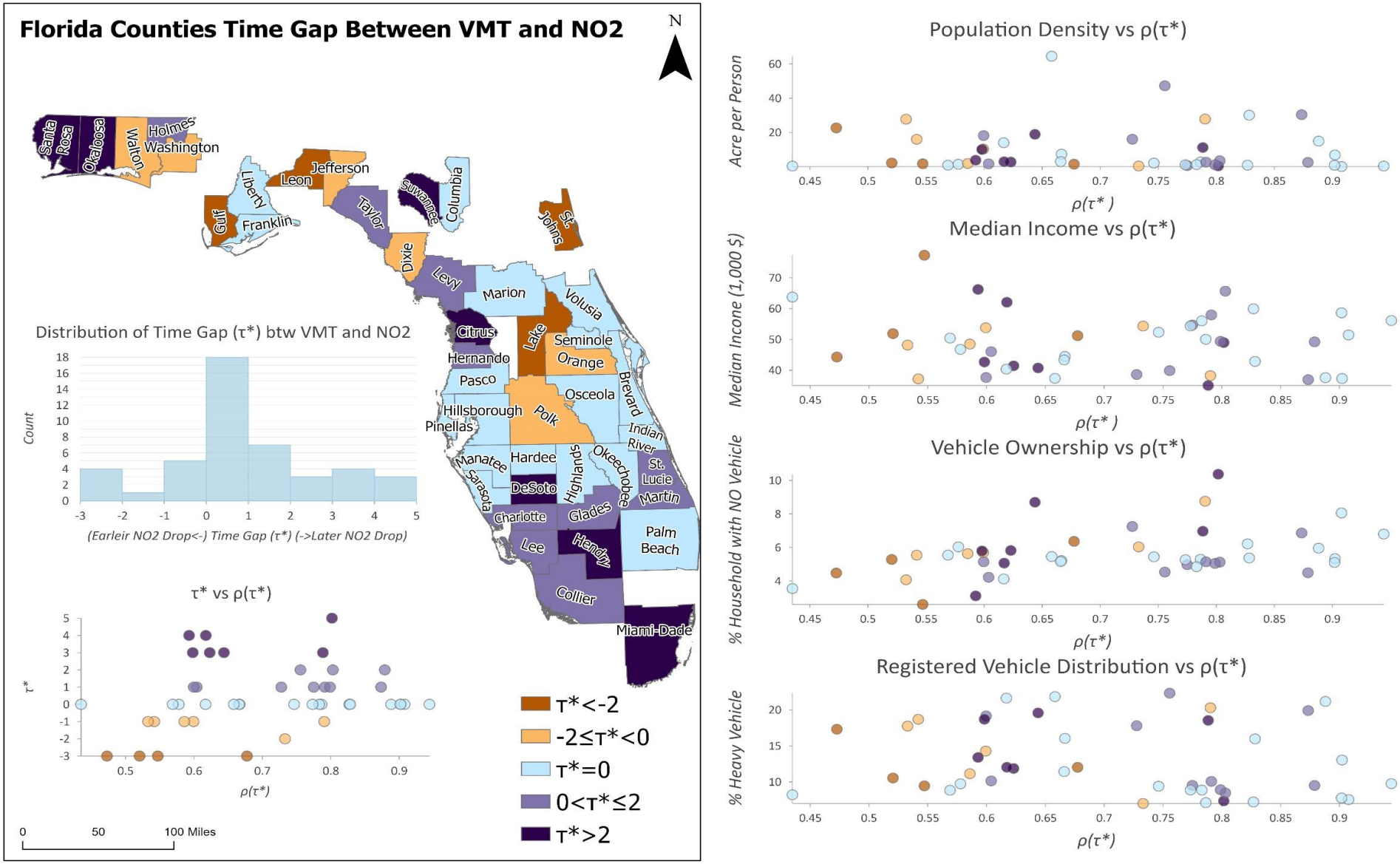


FIGURE 9: Distribution of the time lag between VMT and NO_2 density among Florida counties and potential factors that can affect the correlation between VMT and NO_2 density

CONCLUSION

This study focuses on the impact of COVID-19 both on traffic and NO_2 densities investigating the relationship between the declines on traffic and NO_2 density in all the Florida Counties through March 2020. County level daily NO_2 levels are extracted from Sentinel-5P satellite, which expected to change the way the air quality is monitored today and Vehicle Mile Traveled (VMT) estimates were acquired from cell phone mobility records. Population density, median income and the percentage of the households with no vehicle are also obtained for each county from the US Census's 2018 American Community Survey. Cross-correlation-based dependency analysis was conducted with the Pearson's r statistic to analyze the similarities and associated time lag between 7-day moving averages of VMT and NO_2 density of 67 counties.

Among the 67 Florida Counties, an average of 72.93% and 69.30% decrease were observed for NO_2 and VMT, respectively. This decrease is higher compared to the previous studies since it indicates the percentage difference between the observed maximum and minimum values within the month of March 2020. Majority of the counties have no time lag between VMT and NO_2 density; however, a cluster of South Florida counties presents earlier decrease on NO_2 density compare to VMT, which can be attributed for a decrease in the industrial activities before the traffic reduction. Also, COVID-19 impacts on traffic and NO_2 density have started in the first half of the March 2020, earlier than the official stay-at-home orders. Cross correlation analysis results revealed that the VMT and NO_2 are highly correlated and most of the counties observe similar patterns. The impact of population and economic status of the counties have also been tested regarding the relationship between VMT and NO_2 density; however, no significant impact was observed.

Future studies shall include the weather and commercial industry information from each county so that additional impacts on the relationship between VMT and NO_2 density can be tested. Especially since the NO_2 production can also occur due to commercial manufacturing and food manufacturing, such information can help assess the time lags that occur in different counties. The proposed analysis can also be utilized if other pollutant data sets can be obtained for CO, CO_2, and particular matters (PM). In addition, these results are very site specific, and therefore it would be interesting to study other states that had different experiences with COVID-19.

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AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: AK, NB, MG, RA, EEO data collection: AK, NB, MG analysis and interpretation of results: AK, NB, MG, RA, EEO draft manuscript preparation: AK, NB, RA, EEO. All authors reviewed the results and approved the final version of the manuscript.

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